DESCRIPTIVE BRIEF: THE 2026 DROUGHT IN GUATEMALA

Orlando Valladares

Replication package: github.com/orlando-valladares/guatemala-drought-replication

INTRODUCTION

Guatemala is arguably experiencing its most severe drought in the modern climatological record, both through September and, critically, for its main agricultural season, *Primera* (May–August; Figure 1). Focusing on *Primera*, the CHIRPS dataset recorded a national average rainfall total of 757.1 mm, 1.92 standard deviations below its 1,102 mm historical mean—the second-lowest total since 1981. The ERA5-Land dataset recorded 574.5 mm, 2.49 standard deviations below its 906.0 mm historical mean, the lowest total since 1950. Across both datasets, this is the drought with the highest share of agricultural workers exposed to extreme drought ($z < -2$) during *Primera*, encompassing 54.6% of the census's agricultural workers in CHIRPS and 60.1% in ERA5-Land (Figure 3). Extreme drought ($z < -2$) is a point at which Oxfam's rapid assessment reports a prevalence of nearly complete crop loss for family farmers, and one which the datasets' next-highest worker-exposure years had seen for only 28.3% (2015) and 22.3% of agricultural workers, respectively (Oxfam, 2026). The strongest deficits are concentrated in central, southern, and eastern Guatemala (Figure 4).

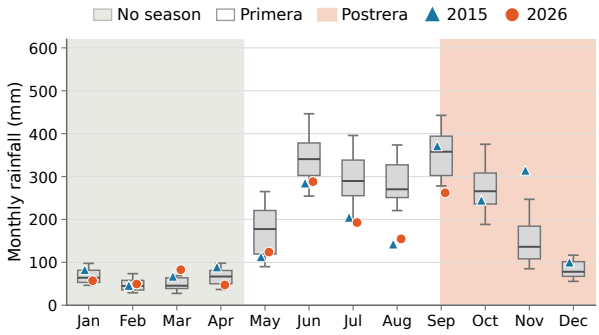
Though alarms about the current drought are increasingly sounded, particularly amid a context of El Niño and climate change, its magnitude is less frequently quantified. This brief contributes to its quantification using datasets that extend into September 2026 and compares it with the well-documented impacts of the 2015 drought, such as crop loss, increases in maize prices as households become dependent on markets for it (wholesale maize, January–peak: +10.4% in 2015; +45.5% in 2026; MAGA, 2015, 2026), and reductions in rural day-labour availability, or *jornaleo*. While final impacts will depend on poverty, resilience, and land-quality factors outside the scope of this brief, the descriptive quantitative evidence presented here suggests a severe multidimensional shock for hundreds of thousands of Guatemala's poor.

DATA AND METHOD

This analysis uses two updated geospatial precipitation datasets, CHIRPS v3 and ERA5-Land, prioritizing the former for its stronger observational basis while using the latter to assess the robustness of the findings. For the main analysis, CHIRPS v3 combines thermal-infrared satellite data with *in situ* precipitation observations to produce a granular rainfall surface on a 0.05-degree grid beginning in 1981 (Funk et al., 2015). For robustness, ERA5-Land, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), provides a model-based reanalysis of land-surface conditions beginning in 1950, distributed on a 0.1-degree grid. For every Guatemalan municipality (ADM2) and department (ADM1), I then aggregate rainfall across intersecting cells and calculate an area-weighted mean in an equal-area projection.

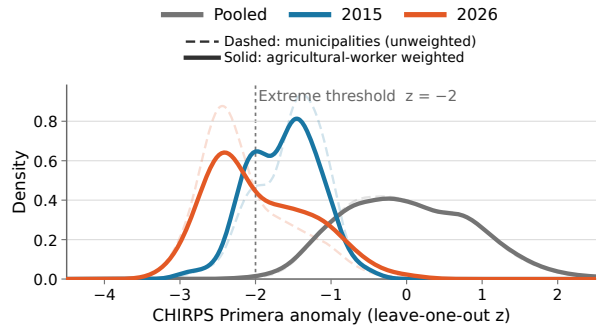
To reduce noise, I aggregate rainfall over the 24 CHIRPS pentads and 123 calendar days in ERA5-Land corresponding to May–August, the period used to approximate *Primera*, the country's main agricultural cycle. In a typical year and location, planting begins with the May–June rains, and harvesting occurs between August and September. Households mainly plant white maize as a monocrop or together with beans and sorghum. A rainfall shock

Figure 1. Guatemala. CHIRPS Monthly rainfall: 2026, 2015, and historical distribution.



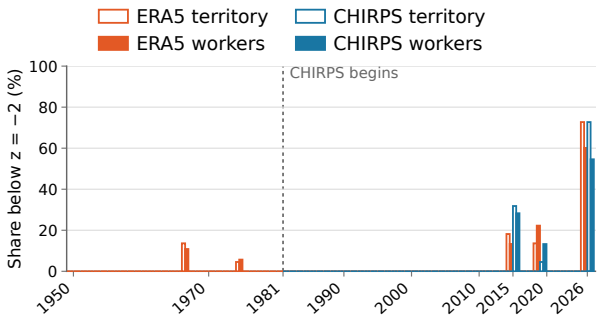
Note. CHIRPS v3 national area-weighted rainfall; final 2026 pentads through September. The September box is split at its midpoint to mark the visual transition from Primera to Postrera.

Figure 2. Guatemala. CHIRPS Distribution of the rainfall anomaly.



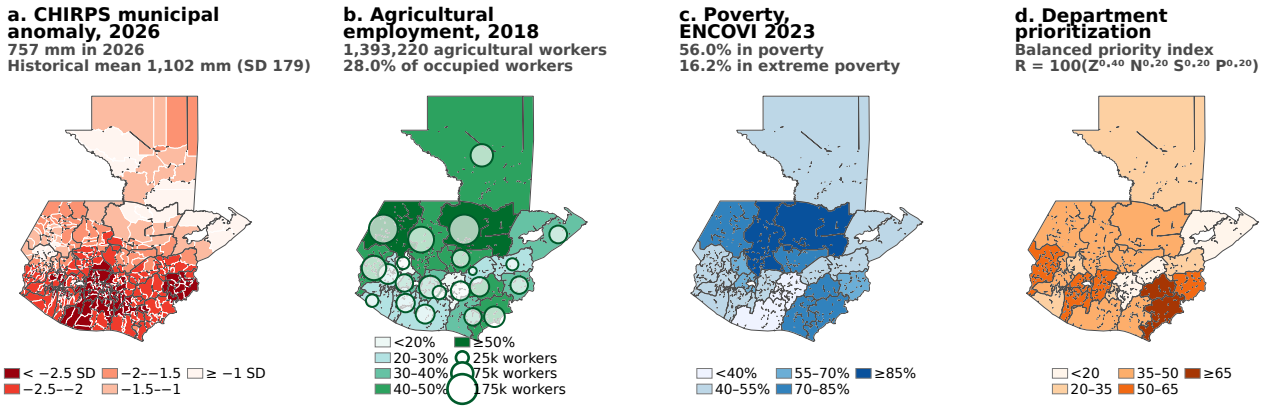
Note. CHIRPS Primera leave-one-out anomalies. Thick lines weight agricultural workers; light lines show the unweighted territorial distribution

Figure 3. Guatemala. Share of territorial units and agricultural workers exposed to extreme drought ($z < -2$), 1950–2026.



Note. ERA5-Land uses 22 departments from 1950; CHIRPS uses 22 departments from 1981. Solid bars show workers and outlined bars show territorial units, applying the fixed 2018 Census worker distribution.

Figure 4. Guatemala. Drought, agricultural employment, poverty, and department prioritization.



Note. Panel a maps municipal CHIRPS Primera anomalies. Panels b–d use departments. Agricultural fill is the share of occupied workers in agriculture; proportional circles show worker counts. Poverty is ENCOVI 2023. Priority uses the balanced index reported in Table A3.

during *Primera* compresses or eliminates the window in which maize, the foundation of the regional diet and a water-intensive crop requiring roughly 500–700 mm of rainfall, can complete its vegetative cycle (Reyna-Méndez, 2018, p. 10). Although rain returns around September and some larger farmers can plant again during *Postrera*, households without irrigation or surplus seed often reduce planted area and prioritize beans or sorghum for their shorter growing cycles or lower seasonal water requirements (SMC/MAGA, 2025, p. 11; Giraldo et al., 2023). To estimate agricultural exposure, I combine CHIRPS with Guatemala's 2018 XII National Population Census (INE, 2019); specifically, the number of workers classified in industry A (agriculture, livestock, forestry, and fishing), which totals 1,393,220 nationally, or 28.0% of the tabulated workforce. The ideal category for this risk analysis would isolate smallholder farmers, but no contemporary official measure identifies them equivalently nationwide. As historical context, Gómez et al. (2013) estimate that small family farmers represented about 62% of agricultural workers in the 2002 Census. Departmental poverty from Guatemala's 2023 household survey, ENCOVI, and other robustness indicators appear in Appendix Table A3 (INE, 2024).

RESULTS

After an already dry 2025 season (865 mm, $z = -1.32$), Guatemala accumulated 757.1 mm of rainfall during *Primera* in 2026, 1.92 standard deviations below its historical mean, while ERA5-Land recorded 574.5 mm ($z = -2.49$, historical mean = 906.0 mm), the lowest total since 1950. Across both datasets, this is the drought with the highest share of agricultural workers exposed to extreme drought ($z < -2$), 54.6% (760,534) and 60.1% (837,965), respectively. By extreme drought, defined here as $z < -2$, I refer to the Standardized Precipitation Index severity convention of McKee et al. (1993), also used by the World Meteorological Organization (2012). For CHIRPS, including severe drought ($z \leq -1.5$) raises agricultural-worker exposure to 72.6%; including moderate drought ($z \leq -1$) raises it to 91.2% (Figure 3).

The sharpest deficits extend from Guatemala's central highlands into its Dry Corridor in the east, through departments including Chimaltenango, Jalapa, and Chiquimula, while the north, Caribbean coast, and parts of the west are relatively less affected (Figure 4, panel a). This is a relatively favorable sign given the socioeconomic vulnerability and agricultural dependence of much of the north. Poverty reaches 90.3% in Alta Verapaz ($z = -1.22$), 86.4% in Quiché ($z = -1.84$), and 81.3% in Huehuetenango ($z = -1.76$), where agricultural employment accounts for 52.9%, 44.3%, and 52.2% of workers, respectively. Nevertheless, because the majority of departments and workers are experiencing severe drought, severely affected departments such as Jalapa ($z = -2.39$) and Jutiapa ($z = -2.47$) still combine high poverty (79.9% and 74.0%) with agricultural dependence (49.2% and 41.0%; Appendix Table A3).

HISTORICAL COMPARISON AND $z < -2$

The 2015 drought is a useful benchmark for translating the magnitude of the current shock into its potential human cost[1]. In CHIRPS, it was slightly more severe nationally than the current drought because it was more spatially homogeneous, despite not having the same prevalence of extreme outcomes ($z < -2$). The present drought therefore combines a similarly low national total with much wider extreme exposure and arrives after greater accumulated drought in its three preceding years: the mean z-score for 2023–2025 was −0.718, compared with +0.575 for 2012–2014 in CHIRPS.

For 2015, the Ministry of Agriculture (MAGA) and SESAN estimated that 154,000–182,000 smallholder households lost 50–100% of their *Primera* maize and bean harvests, reporting losses above 80% in most cases (FAO, 2015; MAGA, 2015). At a macro level, direct maize losses reached Q299.8 million (US$39.3 million), and although

[1]The 2019 drought also ranks among the most severe in both datasets, but its aftermath coincided with the onset of COVID-19 restrictions in 2020, limiting field assessments.

the drought also affected northern regions, 75% of monetary losses were concentrated in five departments, four in the eastern Dry Corridor (MAGA, 2015). While in 2015 these five departments had an average z-score of −2.03 (523 mm), it is now −2.42 (453 mm).

For agricultural households in the hardest-hit areas, the shock combined three mechanisms: near-total crop loss; a seasonal increase in maize prices just as households became market-dependent for food; and a fall in paid agricultural day labour. Less impacted areas suffered a still consequential mix of these. Besides the prevalence of smallholder households losing 80% of their crops, wholesale white-maize prices rose 10.4% to Q149.30/cwt from January to July 2015, and MAGA reported community-market increases of 33.3% in the Dry Corridor and 21.7% nationally (MAGA, 2015). Third, FEWS NET field visits reported that day-labour availability fell 50% in the Dry Corridor and the west and that the household income typically derived from it contracted 36.8% (FEWS NET, 2015, 2016). In the worst cases, 56.4% of surveyed agricultural households in the eastern Dry Corridor and 41.5% in the western highlands consumed seed reserved for the next harvest, prolonging their food vulnerability and market dependence (OXFAM 2015).

CONCLUSION: CURRENT PICTURE

The three mechanisms observed in 2015 may be converging again in 2026. Oxfam's August 2026 rapid assessment in the Central American Dry Corridor suggests a high prevalence of 75–100% crop losses in the now-larger share of areas under extreme drought. Across the 15 departments in its regional sample (4,260 participants in 354 focus groups), 94.9% of community households in the eight departments under extreme drought reported losses of 75–100%, compared with 53.6% in the other seven. In Guatemala's six sampled departments (1,734 participants in 157 focus groups), an average of 98.8% in Jalapa ($z = -2.39$) and Chiquimula ($z = -2.69$) reported losses of 75–100%; in Quiché, Huehuetenango, and Alta Verapaz (average $z = -1.6$), the corresponding average was 44.4%, and in Baja Verapaz it was 86.7% ($z = -1.88$) (Oxfam, 2026). The contrast is suggestive rather than representative because the assessment prioritised historically vulnerable locations. Meanwhile, wholesale white-maize prices rose 45.5% from January to September (MAGA, 2026), while Oxfam reports reductions of at least 50% in day-labour opportunities among 79.4% of affected households in its Guatemalan sample (Oxfam, 2026). If 2015 devastated 154,000 smallholder households and their maize and bean crops, with just 28.3% of agricultural workers exposed to extreme drought, what will 2026, with exposure nearly doubled to 54.6%, mean for Guatemala's poor?

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Table A3. Drought, agricultural exposure, poverty, and robustness indicators, Guatemala

	Hydrological shock			Agricultural exposure		Poverty (ENCOVI 2023)		Balanced index: $R = 100(Z^{.40}N^{.20}S^{.20}P^{.20})$					Robustness			
Unit	2026 rainfall (mm)	Historical mean (SD)	z CHIRPS	Agricultural workers (N)	Agricultural share (%)	Overall (%)	Extreme (%)	$Z^{.40}$	$N^{.20}$	$S^{.20}$	$P^{.20}$		Oxfam 75–100%	SOMIS NDVI Sep	Temp. 2026 (hist.)	
National	**757**	**1102 (179)**	**−1,92**	**1 393 220**	**28,0**	**56,0**	**16,2**	**–**	**–**	**–**	**–**		**–**	**0,17**	**2,49**	**24,4 (22,9)**
Jutiapa	512	890 (153)	−2,47	64 206	41,0	74,0	18,8	65,9	70,5	70,5	70,5		–	−0,98	3,29	24,9 (22,6)
Jalapa	387	772 (161)	−2,39	55 551	49,2	79,9	34,5	52,3	61,4	88,6	79,5		100,0	−3,29	2,93	21,5 (20,0)
Chimaltenango	507	1051 (191)	−2,85	79 578	37,2	47,8	5,8	93,2	79,5	56,8	25,0		–	−2,61	1,58	18,8 (17,5)
Chiquimula	375	821 (165)	−2,69	44 231	38,5	59,6	25,3	84,1	34,1	61,4	56,8		97,6	−1,23	3,03	23,3 (22,0)
Sololá	581	1133 (220)	−2,51	44 230	31,5	61,9	8,8	75,0	29,5	43,2	61,4		–	1,20	3,39	17,3 (16,2)
Suchitepéquez	828	1588 (307)	−2,48	49 589	28,1	53,7	7,7	70,5	47,7	34,1	47,7		–	1,87	3,03	26,1 (24,0)
San Marcos	923	1333 (194)	−2,11	99 248	40,0	53,7	6,9	34,1	84,1	65,9	52,3		–	1,30	2,92	19,5 (18,0)
Santa Rosa	611	1115 (209)	−2,41	44 063	33,2	74,0	16,2	56,8	25,0	47,7	75,0		–	0,35	−2,57	24,4 (22,5)
Quiché	898	1373 (258)	−1,84	115 896	44,3	86,4	34,4	20,5	88,6	75,0	93,2		53,5	0,40	−1,79	20,2 (19,1)
Huehuetenango	783	1221 (249)	−1,76	161 393	52,2	81,3	29,7	15,9	93,2	93,2	88,6		40,8	0,35	−1,44	18,9 (17,6)
Baja Verapaz	472	794 (172)	−1,88	45 338	46,7	80,1	35,8	25,0	38,6	79,5	84,1		86,8	−1,48	−2,40	21,2 (19,9)
Alta Verapaz	1213	1551 (276)	−1,22	185 033	52,9	90,3	58,5	11,4	97,7	97,7	97,7		39,0	1,72	−1,92	24,3 (23,0)
Quetzaltenango	823	1353 (218)	−2,43	58 843	21,8	44,1	2,8	61,4	65,9	20,5	15,9		–	0,67	−3,05	20,2 (18,8)
Escuintla	677	1336 (248)	−2,65	52 812	21,8	39,2	2,6	79,5	52,3	15,9	11,4		–	1,61	−3,25	27,1 (24,8)
Retalhuleu	674	1228 (235)	−2,36	24 504	24,4	50,9	4,1	47,7	15,9	25,0	38,6		–	0,98	−2,95	27,6 (25,5)
Zacapa	434	823 (178)	−2,19	23 357	28,4	52,6	13,5	38,6	6,8	38,6	43,2		–	−2,35	−1,53	23,3 (22,2)
Sacatepéquez	387	853 (158)	−2,95	27 536	19,5	38,7	2,5	97,7	20,5	11,4	6,8		–	−1,87	−2,77	19,5 (17,6)
Petén	700	904 (177)	−1,16	79 433	48,1	50,6	8,4	6,8	75,0	84,1	34,1		–	−0,19	−2,07	27,9 (26,4)
Totonicapán	342	613 (131)	−2,07	23 983	17,5	62,1	10,1	29,5	11,4	6,8	65,9		–	−2,37	−1,73	16,2 (15,0)
El Progreso	315	671 (161)	−2,20	15 027	24,8	46,4	5,1	43,2	2,3	29,5	20,5		–	−1,76	−2,06	22,8 (21,4)
Guatemala	395	833 (161)	−2,72	53 776	4,1	21,6	1,1	88,6	56,8	2,3	2,3		–	−2,84	−2,13	21,5 (19,7)
Izabal	1188	1311 (200)	−0,62	45 593	33,7	50,5	14,9	2,3	43,2	52,3	29,5		–	1,66	−1,06	26,1 (25,2)

Note. CHIRPS v3 rainfall is accumulated over *Primera* (May–August); its historical mean, SD, and z-score use 1981–2025 separately for each geography. Agricultural-worker counts and shares are industry A of the 2018 Census. Overall and extreme poverty are person-weighted FGT0 incidence from ENCOVI 2023. The balanced index is the weighted geometric mean of mid-rank percentiles across 22 departments: drought severity Z (40%), agricultural workers N (20%), agricultural employment share S (20%), and overall poverty P (20%); it is a relative prioritization score, not a probability or impact estimate. Oxfam reports the percentage of households reporting 75–100% crop losses in purposively selected communities; –– denotes departments outside that non-representative assessment. For MODIS, MAGA 2025 Level-1 *Agricultural territories* polygons were reprojected and rasterized at four times the MOD13A3.061 grid resolution to retain each roughly 1-km pixel's fractional agricultural coverage. That fixed mask weights the August–September NDVI mean in every year; pixels must have valid NDVI and SummaryQA 0 or 1 in both months. The 2026 z-score uses the geography-specific mean and sample SD of the 25 annual August–September values in 2001–2025. The mapped mask is not planted area, and NDVI is not crop loss or yield. ERA5-Land rainfall z and historical temperature mean are calculated separately for each geography over the full 1950–2025 record; historical temperature appears in parentheses. Both are alternative robustness measures.

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